CYBER AND NETWORK SECURITY

FIVE MAJOR CYBERATTACKS

Detailed article-style visuals • case-study analysis • definitions • prevention lessons

A second edition featuring Target, WannaCry and the NHS, Capital One, MOVEit and Change Healthcare - five documented incidents with different attack methods and consequences. Each case includes a large article-style evidence page followed by a separate analysis page with definition, timeline, impact, response, security lessons and live source links.

Editorial note: Article visuals are original educational reconstructions based on the cited reporting. They are not literal publisher screenshots or scans of original newspapers. The report summarises source information in its own wording and links to the full source for verification.

How to use this report

Each case opens with a full-page newspaper-style visual showing the publisher, headline, date, key figures and a detailed summary. The following page defines the attack, sets out the timeline, impact and response, notes how reliable the sources are, and lists practical prevention lessons.

Five cases selected

1. Target payment-card data breach - settlement announced 23 May 2017 (breach in 2013)
2. WannaCry ransomware attack on the NHS - NAO report 27 October 2017 (attack 12 May 2017)
3. Capital One cloud data theft - 29 July 2019
4. MOVEit Transfer zero-day exploitation - 7 June 2023
5. Change Healthcare ransomware attack - congressional hearing 1 May 2024 (attack in February 2024)

Key terms

Data breach: unauthorised access to or disclosure of protected information.

Ransomware: malicious software used to block access to systems or information, often to demand payment.

Zero-day vulnerability: a software flaw exploited before the vendor has released a fix.

Multi-factor authentication (MFA): a login that needs a second proof of identity beyond a password.

Cloud misconfiguration: an incorrect setting or permission in a cloud service that exposes systems or data.

Third-party risk: risk created when a supplier, vendor or product has access to, or is trusted by, your systems.

Evidence note: Regulator statements, court records, company testimony and media reporting can describe different stages of an investigation. This report names the type of source behind each figure, dates figures that changed over time, and does not treat unverified entry-path claims as established facts.

ATTORNEY GENERAL OF CONNECTICUT

CYBERSECURITY CASE 01

TARGET / NEWS AND OFFICIAL REPORTING

47 States and D.C. Settle with Target for $18.5 Million over 2013 Data Breach

Attackers used credentials from a third-party vendor to enter Target's network, then captured payment-card and customer data during the pre-Christmas shopping period.

BY STATE ATTORNEYS GENERAL (MULTISTATE PRESS RELEASE)

PUBLICATION / ALERT DATE: 23 MAY 2017

41 MILLION
payment-card accounts affected in the 2013 breach

60 MILLION+
customers whose contact information was exposed

$18.5M
multistate settlement, the largest of its kind at the time

THE BREACH

Between 27 November and 15 December 2013, intruders reached Target's systems and stole customer data in the middle of the pre-Christmas shopping period. State attorneys general later said the incident affected more than 41 million payment-card accounts and exposed contact details for over 60 million customers. Target overhauled its security and offered free credit reports to shoppers.

HOW ATTACKERS GOT IN

According to the attorneys general, the attackers first obtained login credentials belonging to a third-party HVAC (heating and cooling) vendor. Those credentials were used to reach a gateway server and exploit weaknesses in Target's network, which led to a customer-service database and let the intruders install malware that captured data.

WHY THE DAMAGE SPREAD

A vendor login should not be a path to systems that handle payment data. The settlement terms point to the controls regulators wanted improved: separating the cardholder data environment from the rest of the network, maintaining and supporting software, tighter access control and two-factor authentication for certain accounts.

WHAT INFORMATION WAS EXPOSED

The attorneys general said the stolen data included full names, telephone numbers, email and mailing addresses, payment-card numbers with expiry dates and CVV1 codes, and encrypted debit PINs. Card details like these can be used for fraud until the cards are cancelled and replaced, so this type of breach creates immediate financial risk.

THE REGULATORY RESPONSE

On 23 May 2017, 47 states and the District of Columbia announced an $18.5 million settlement, described at the time as the largest multistate data-breach settlement. Target also had to commission an independent security assessment and adopt encryption, software-maintenance, segmentation and access-control measures. Officials noted that Target cooperated with the investigation.

THE LONG-TERM LESSON

Suppliers are part of an organisation's attack surface. Vendor accounts need least-privilege access limited to the systems they truly need, strong authentication and monitoring. Payment systems should sit behind separate network boundaries so that one stolen password cannot reach everything.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 1 / ANALYSIS

Target data breach

Incident / disclosure: 23 May 2017 (settlement); breach in Nov-Dec 2013 Case: TARGET

Definition of the attack

A third-party (supply-chain access) compromise happens when attackers use the credentials or connection of a supplier to enter a customer's network. In the Target case, stolen vendor credentials were reported to be the first step, followed by malware that captured payment-card data. The result was a large data breach.

IMPACT

Payment-card data for more than 41 million accounts and contact details for more than 60 million customers were exposed, creating fraud risk, card-replacement costs and loss of consumer trust.

RESPONSE

Target investigated, overhauled its security and offered free credit reports. A multistate investigation led by state attorneys general ended in an $18.5 million settlement that required security improvements, including an independent assessment.

TIMELINE

27 Nov - 15 Dec 2013 - Window in which the breach took place, according to the states.
Dec 2013 - Breach reported publicly; free credit reports offered to shoppers.
23 May 2017 - 47 states and D.C. announce the $18.5 million settlement.

SOURCE NOTE

The settlement agreement (an Assurance of Voluntary Compliance) did not set out factual allegations about the breach. The entry path described here comes from state attorney-general press releases, so it is presented as the states' account and not as a court finding.

Prevention and lessons learned

- Give vendors least-privilege accounts limited to the systems they need.
- Require multi-factor authentication for remote and third-party access.
- Segment payment-card systems from the general corporate network.
- Monitor for unusual logins and unexpected software on servers that handle card data.
- Review supplier security as part of regular risk assessments and contracts.

Primary source:
https://portal.ct.gov/ag/press-releases-archived/2017-press-releases/ag-jepsen-conn-leads-185m-settlement-with-target-corporation-over-2013-data-breach

Additional source: https://www.mass.gov/news/in-largest-settlement-of-its-kind-massachusetts-to-receive-625000-after-investigation-into-target-data-breach

NATIONAL AUDIT OFFICE (UK)

CYBERSECURITY CASE 02

WANNACRY / NHS / NEWS AND OFFICIAL REPORTING

Auditor: NHS Could Have Prevented WannaCry by Following Basic IT Security Practice

A fast-spreading ransomware outbreak locked devices across NHS organisations in England, disrupted patient care and exposed ageing systems and untested emergency plans.

BY THE NATIONAL AUDIT OFFICE

PUBLICATION / ALERT DATE: 27 OCTOBER 2017

12 MAY 2017
day WannaCry began locking NHS devices

81 of 236
hospital trusts in England disrupted

603
other NHS bodies infected, including primary care

THE ATTACK

On Friday 12 May 2017 a global ransomware attack known as WannaCry reached the NHS in England. The Department of Health declared a major incident. That evening a security researcher activated a 'kill switch' that stopped WannaCry from locking more devices. The National Audit Office called it the largest cyber attack to affect the NHS.

HOW IT SPREAD

WannaCry behaved like a worm: it spread between connected computers automatically, so one infected machine could affect many others. The NAO described it as relatively unsophisticated and preventable with basic IT security, such as moving off old software and keeping internet-facing firewalls properly secured.

WHAT WARNINGS WERE MISSED

The Department and Cabinet Office told trusts in 2014 to have robust plans to leave unsupported software such as Windows XP by April 2015. The Department was also warned about cyber risks a year before the attack, yet did not formally respond until July 2017 and did not know how prepared local organisations were.

THE IMPACT

At least 81 of 236 trusts were disrupted and 603 other NHS bodies, including GP practices, were infected. The NAO reported at least 19,000 appointments cancelled. NHS Digital believed no patient data was stolen, and the Department could not say how much the disruption cost.

THE RESPONSE

Because the NHS had never rehearsed a national cyber attack, it was unclear who should lead the response. Email was unavailable at some sites, so staff used personal mobile devices, including WhatsApp, to coordinate. Many organisations also reported the incident to different bodies, including local police.

THE LONG-TERM LESSON

An emergency plan only works if it has been tested locally as well as nationally. Organisations need a live inventory of unsupported systems, a patching programme with clear owners, tested backups and a communication method that still works when email and networks are down.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 2 / ANALYSIS

WannaCry ransomware attack on the NHS

Incident / disclosure: 27 October 2017 (NAO report); attack on 12 May 2017 Case: WANNACRY / NHS

Definition of the attack

A ransomware worm is malicious software that locks files or devices to demand payment and also spreads automatically across networks without anyone clicking a link. WannaCry spread by exploiting a known weakness in older, unpatched Windows systems.

IMPACT

Services in the week from 12 to 19 May 2017 were affected: at least 81 trusts were disrupted, 603 other NHS bodies were infected and at least 19,000 appointments were cancelled.

RESPONSE

The Department declared a major incident and national bodies coordinated the response. A researcher's kill switch stopped further locking. The NAO then recommended tested response plans, clear roles and better IT security assurance.

TIMELINE

2014 - Government tells trusts to plan to migrate away from Windows XP by April 2015.
12 May 2017 - WannaCry locks devices; major incident declared; kill switch used that evening.
12-19 May 2017 - Period in which NHS services were affected.
27 Oct 2017 - NAO publishes its investigation.

SOURCE NOTE

The NAO covered events up to 30 September 2017 and only the NHS in England. The view that no patient data was stolen is NHS Digital's, as reported by the NAO. The total cost of the disruption was not established.

Prevention and lessons learned

- Keep an inventory of operating systems and retire unsupported ones by a set deadline.
- Apply critical patches quickly and verify that they were installed.
- Secure internet-facing firewalls and block unnecessary network paths between devices.
- Rehearse the incident plan locally and nationally, including who leads.
- Prepare backup communication channels and offline fallback procedures for clinical work.

Primary source:
https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/

Additional source: https://www.nao.org.uk/wp-content/uploads/2017/10/Investigation-WannaCry-cyber-attack-and-the-NHS-Summary.pdf

CAPITAL ONE / NEWS AND OFFICIAL REPORTING

Seattle Tech Worker Arrested over Data Theft from Capital One

A misconfigured web application firewall in a cloud environment let an intruder reach stored credit-application data belonging to more than 100 million people.

BY THE U.S. DEPARTMENT OF JUSTICE

PUBLICATION / ALERT DATE: 29 JULY 2019

~100M + 6M
people affected in the United States and Canada

17 JUL 2019
tip from a GitHub user alerted the bank

$80M
civil penalty imposed by the OCC in August 2020

THE THEFT

Capital One said an outside individual gained unauthorised access to personal information from credit-card applications. Federal prosecutors charged a former Seattle software engineer, Paige Thompson, with computer fraud and abuse. The company said about 100 million people in the United States and about 6 million in Canada were affected.

HOW IT WAS DETECTED

According to the criminal complaint, the suspect posted about the theft on GitHub. On 17 July 2019 a GitHub user alerted Capital One. The bank confirmed an intrusion on 19 July and contacted the FBI, which traced the activity. The arrest followed on 29 July.

HOW THE INTRUSION WORKED

The Department of Justice said the intrusion occurred through a misconfigured web application firewall that enabled access to stored data. The OCC later found that, after moving IT operations to the cloud around 2015, the bank lacked effective risk assessment, network security controls, data-loss prevention and alert handling.

WHAT DATA WAS INVOLVED

The exposed records included names, addresses, phone numbers, credit scores and payment histories from card applications. A smaller number of records included Social Security numbers and linked bank account numbers. Capital One reported that the FBI recovered the data and that it saw no evidence it had been used for fraud or shared.

LEGAL AND REGULATORY OUTCOME

A grand jury indicted Thompson in August 2019 on wire fraud and computer data theft charges, and she was convicted of wire fraud and computer intrusion offences in June 2022. In August 2020 the OCC fined the bank $80 million and required a plan to improve cloud risk management, while crediting its customer notification efforts.

THE LONG-TERM LESSON

Moving to the cloud does not move responsibility away from the customer. Firewalls, access roles and storage must be configured correctly, reviewed regularly and tested, with alerts that someone investigates and audit findings that reach the board.

RECONSTRUCTED ARTICLE VISUAL
Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 3 / ANALYSIS

Capital One cloud data theft

Incident / disclosure: 29 July 2019 Case: CAPITAL ONE

Definition of the attack

A cloud misconfiguration is an incorrect setting or permission in a cloud environment that leaves systems or data reachable by someone who should not have access. In this case, prosecutors identified a misconfigured web application firewall as the weakness the intruder used.

IMPACT

Personal and financial application data for roughly 106 million people was exposed, leading to credit-monitoring offers, lawsuits, regulatory action and response costs the company expected to be about $100-150 million.

RESPONSE

Capital One contacted the FBI after confirming the intrusion and offered free credit monitoring and identity protection. The OCC later issued consent orders and an $80 million penalty requiring stronger cloud risk management.

TIMELINE

2015 - Capital One moves a major part of its IT operations to a public cloud (OCC finding).
Mar - Jul 2019 - Period of the intrusion, according to later reporting.
17-19 Jul 2019 - GitHub tip received; intrusion confirmed; FBI contacted.
29 Jul 2019 - Arrest and public disclosure.
6 Aug 2020 - OCC announces $80 million civil penalty.
Jun 2022 - Conviction in federal court.

SOURCE NOTE

The July 2019 DOJ release summarises a criminal complaint, which contains allegations. The OCC's findings come from a consent order the bank agreed to without admitting or denying them. The 2022 conviction was announced separately by the DOJ.

Prevention and lessons learned

- Assess cybersecurity risk before moving major systems to the cloud.
- Review firewall rules, roles and storage permissions with automated configuration checks.
- Use data-loss prevention and alerting, and investigate alerts promptly.
- Report control weaknesses found by internal audit to senior management and the board.
- Publish a clear route for researchers and the public to report exposed data.

Primary source:
https://www.justice.gov/usao-wdwa/pr/seattle-tech-worker-arrested-data-theft-involving-large-financial-services-company

Additional source (indictment): https://www.justice.gov/usao-wdwa/pr/former-seattle-tech-worker-indicted-federal-charges-wire-fraud-and-computer-data-theft

Regulator source (OCC consent order): https://www.occ.gov/static/enforcement-actions/ea2020-036.pdf

MOVEIT / NEWS AND OFFICIAL REPORTING

#StopRansomware: CL0P Gang Exploits MOVEit Transfer Vulnerability

A previously unknown flaw in widely used file-transfer software let attackers steal data from the databases behind many organisations' MOVEit servers.

BY CISA AND THE FBI

PUBLICATION / ALERT DATE: 7 JUNE 2023

27 MAY 2023
earliest exploitation noted in the advisory

ZERO-DAY
CVE-2023-34362, a SQL injection flaw

7 JUN 2023
CISA and FBI publish a joint advisory

THE ADVISORY

On 7 June 2023 the FBI and CISA warned that the CL0P ransomware gang, also known as TA505, was exploiting a vulnerability in Progress Software's MOVEit Transfer, a managed file-transfer product used to move sensitive files. Open-source information cited in the advisory put the first exploitation on 27 May 2023.

HOW THE ATTACK WORKED

The flaw, CVE-2023-34362, is a SQL injection weakness that was unknown to the vendor when attackers began using it. Internet-facing MOVEit Transfer web applications were infected with a web shell called LEMURLOOT, which gave the attackers a foothold and let them steal data from the underlying databases.

WHY FILE-TRANSFER TOOLS ARE TARGETED

File-transfer servers hold sensitive files from many business partners in one place and are often reachable from the internet. One flaw therefore gives attackers a route to many victims at once, without having to break into each organisation separately.

A REPEATED PATTERN

The advisory noted that TA505 had earlier used zero-day exploits against Accellion FTA servers in 2020-2021 and against the GoAnywhere MFT platform in January 2023. The group repeatedly chose file-transfer products as targets.

DATA THEFT AS THE GOAL

In this campaign the main objective was stealing data rather than encrypting systems, so good backups alone would not remove the extortion risk. Defence against ransomware groups therefore has to cover data theft and extortion as well as encryption.

THE LONG-TERM LESSON

Internet-facing software that stores sensitive files deserves the same care as a database. Keep an inventory of such tools, patch urgently, check for web shells and unusual outbound transfers after fixing, and avoid keeping files on transfer servers longer than needed.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 4 / ANALYSIS

MOVEit Transfer zero-day exploitation

Incident / disclosure: 7 June 2023 Case: MOVEIT

Definition of the attack

A zero-day vulnerability is a software flaw that attackers exploit before the vendor has a fix. A SQL injection is a flaw in which crafted input changes the database commands an application runs. In MOVEit Transfer, this combination allowed attackers to steal data from the databases behind exposed servers.

IMPACT

Data held in MOVEit Transfer databases was stolen from affected organisations. Because many organisations used the same product, one vulnerability created many victims at the same time and complicated notification and investigation.

RESPONSE

Progress disclosed the flaw and released fixes. The FBI and CISA published indicators of compromise, tactics and mitigations. Organisations had to patch, check for the LEMURLOOT web shell and work out what data had been taken.

TIMELINE

Jan 2023 - CL0P zero-day campaign against GoAnywhere MFT (CVE-2023-0669).
27 May 2023 - Earliest reported exploitation of CVE-2023-34362.
31 May 2023 - Progress discloses the MOVEit Transfer vulnerability.
7 Jun 2023 - CISA and FBI release advisory AA23-158A.

SOURCE NOTE

Exploitation dates come from the advisory, which cites open-source information. The report does not state a total number of victim organisations because published totals changed over time.

Prevention and lessons learned

- Keep an inventory of internet-facing applications, especially file-transfer and remote-access tools.
- Patch emergency vulnerabilities fast, then check for web shells and other persistence.
- Monitor for unusual outbound data transfers from servers that hold files.
- Limit how long sensitive files stay on transfer servers and encrypt them at rest.
- Ask suppliers how they will notify you about zero-day flaws in their products.

Primary source:
https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-158a

Additional source (vendor disclosure dates): https://www.tenable.com/blog/faq-for-MOVEit-Transfer-vulnerabilities-CVE-2023-34362-and-cl0p-ransomware-gang

CHANGE HEALTHCARE / NEWS AND OFFICIAL REPORTING

Hearing: Examining the Change Healthcare Cyberattack

Criminals entered through a remote-access portal that lacked multi-factor authentication, then locked systems that process healthcare claims and payments across the United States.

BY THE U.S. HOUSE OF REPRESENTATIVES

HEARING DATE: 1 MAY 2024

9 DAYS
between first access (12 Feb) and ransomware (21 Feb 2024)

$22M
ransom paid in bitcoin, per the UnitedHealth CEO

192.7M
people, latest figure reported to HHS (July 2025)

THE ATTACK

On 21 February 2024 Change Healthcare, a UnitedHealth Group subsidiary, announced a cyberattack. Systems were disconnected, which disrupted claims, billing, insurance authorisations and pharmacy payments across the country for weeks. The American Medical Association said providers lost more than $100 million a day.

HOW ATTACKERS GOT IN

UnitedHealth's CEO told Congress that criminals used compromised credentials on 12 February to access a remote-access portal that did not have multi-factor authentication. They stayed inside for nine days before deploying ransomware. The group behind the attack was reported to be ALPHV/BlackCat.

WHY IT SPREAD

Lawmakers asked how a login without MFA could lead to such broad access, why the intruders were not detected sooner and why recovery took so long. The CEO said Change, acquired in 2022, was still being upgraded, that MFA was standard elsewhere in the group, and that he was frustrated it was missing.

THE RANSOM

The CEO confirmed that a $22 million ransom was paid in bitcoin and called it one of the hardest decisions he had made. Paying a ransom does not guarantee that stolen data is deleted or that systems return to normal quickly, and it can encourage further attacks.

HOW THE SCALE GREW

Change first said about 100 million people were affected, later estimated about 190 million, and in July 2025 reported 192.7 million to the HHS Office for Civil Rights, making it the largest healthcare data breach in the US. UnitedHealth booked $1.1 billion in total attack costs in the second quarter of 2024.

THE LONG-TERM LESSON

Multi-factor authentication must cover every remote-access path, especially systems inherited through acquisitions. Healthcare also needs network segmentation, tested offline backups and fallback payment processes, because one processor failing can stop services far beyond its own customers.

RECONSTRUCTED ARTICLE VISUAL
Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 5 / ANALYSIS

Change Healthcare ransomware attack

Incident / disclosure: 1 May 2024 (hearing); attack announced 21 February 2024 Case: CHANGE HEALTHCARE

Definition of the attack

A credential-based ransomware attack starts with valid stolen login details instead of a software exploit. Multi-factor authentication (MFA) adds a second proof of identity, such as a one-time code, so a stolen password alone is not enough. Change Healthcare shows what can happen when that second layer is missing.

IMPACT

Claims, billing, eligibility checks and pharmacy payments were disrupted nationwide. Patient data for an estimated 192.7 million people was reported exposed, and providers faced serious cash-flow shortages.

RESPONSE

UnitedHealth disconnected affected systems, rebuilt its network and paid a ransom. By 1 May it had issued $6.5 billion in payments and no-interest loans to providers. The CEO testified to the House and Senate, and HHS OCR was assessing HIPAA compliance (as reported in 2025).

TIMELINE

12 Feb 2024 - Attackers access the remote portal with compromised credentials.
21 Feb 2024 - Ransomware deployed; Change Healthcare announces the attack.
1 May 2024 - CEO testifies to House and Senate committees.
Jan 2025 - Company estimates about 190 million people affected.
31 Jul 2025 - Figure of 192.7 million submitted to HHS OCR.

SOURCE NOTE

Entry-path details come from the CEO's testimony as summarised in the hearing record and news reports. Victim counts changed several times, so each figure is dated. Figures after July 2025 may have been updated again; check HHS OCR.

Prevention and lessons learned

- Enforce MFA on all remote access, including legacy and acquired systems.
- Bring acquired companies' systems into security monitoring early.
- Detect and respond to unusual account activity; nine days of access is too long.
- Keep tested, isolated backups and a rebuild plan for critical services.
- Plan manual fallbacks and publish figures with dates, since estimates change.

Primary source:
https://docs.house.gov/meetings/IF/IF02/20240501/117242/HHRG-118-IF02-MState-M001159-20240501.pdf

Additional source (hearing coverage): https://www.healthcaredive.com/news/change-healthcare-cyberattack-congress-unitedhealth-andrew-witty/714954/

Figure update (192.7 million): https://cyberinsider.com/change-healthcare-revises-breach-impact-to-192-7-million-individuals/

Lessons across all five incidents

Different sectors, different attack paths - but the same weaknesses keep returning.

1. Stolen credentials need a second lock

Target (vendor login) and Change Healthcare (remote portal without MFA) show how one valid login can open a path to millions of records. Use multi-factor authentication, least privilege and monitoring for unusual account behaviour.

2. Patch, retire and watch internet-facing software

WannaCry hit unsupported, unpatched systems, while MOVEit was exploited as a zero-day. Maintain an inventory, retire unsupported software, patch urgently, and check for signs of compromise after fixing.

3. Suppliers and products extend the attack surface

A vendor account at Target and a shared file-transfer product in MOVEit both gave attackers leverage. Review supplier access and know which third-party products are exposed to the internet.

4. Cloud responsibility stays with the customer

Capital One shows that moving to the cloud still requires risk assessment, correct configuration, data-loss prevention, alert handling and board-level oversight.

5. Rehearse recovery and date your estimates

The NHS had not rehearsed a national response, and Change Healthcare struggled to recover. Victim counts also changed (Change Healthcare rose from about 100 million to 192.7 million), so reports should say which source and date each figure comes from.

Summary of the five cases

Case	Year	Attack type	Entry point	Key weakness
Target	2013	Credential theft and card-data malware	Third-party vendor credentials (per attorneys general)	Weak segmentation and access controls
WannaCry / NHS	2017	Ransomware worm	Unsupported and unpatched Windows systems	Legacy software; untested response plan
Capital One	2019	Cloud data theft	Misconfigured web application firewall	Cloud risk assessment and alerting gaps (OCC)
MOVEit	2023	Zero-day exploitation; data-theft extortion	SQL injection in MOVEit Transfer	Internet-facing file-transfer software holding sensitive data
Change Healthcare	2024	Ransomware	Compromised credentials on remote portal	No MFA; nine days of undetected access

Sources and verification

The links below are the primary source material used for the case pages, with additional official or reporting sources where useful. Open the linked sources to consult the full details and wording.

1. Target data breach

Primary source: https://portal.ct.gov/ag/press-releases-archived/2017-press-releases/ag-jepsen-conn-leads-185m-settlement-with-target-corporation-over-2013-data-breach

Additional source: https://www.mass.gov/news/in-largest-settlement-of-its-kind-massachusetts-to-receive-625000-after-investigation-into-target-data-breach

2. WannaCry ransomware attack on the NHS

Primary source: https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/

Additional source: https://www.nao.org.uk/wp-content/uploads/2017/10/Investigation-WannaCry-cyber-attack-and-the-NHS-Summary.pdf

3. Capital One cloud data theft

Primary source: https://www.justice.gov/usao-wdwa/pr/seattle-tech-worker-arrested-data-theft-involving-large-financial-services-company

Additional source (indictment): https://www.justice.gov/usao-wdwa/pr/former-seattle-tech-worker-indicted-federal-charges-wire-fraud-and-computer-data-theft

Regulator source (OCC consent order): https://www.occ.gov/static/enforcement-actions/ea2020-036.pdf

4. MOVEit Transfer zero-day exploitation

Primary source: https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-158a

Additional source (vendor disclosure dates): https://www.tenable.com/blog/faq-for-MOVEit-Transfer-vulnerabilities-CVE-2023-34362-and-cl0p-ransomware-gang

5. Change Healthcare ransomware attack

Primary source: https://docs.house.gov/meetings/IF/IF02/20240501/117242/HHRG-118-IF02-MState-M001159-20240501.pdf

Additional source (hearing coverage): https://www.healthcaredive.com/news/change-healthcare-cyberattack-congress-unitedhealth-andrew-witty/714954/

Figure update (192.7 million): https://cyberinsider.com/change-healthcare-revises-breach-impact-to-192-7-million-individuals/

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